

FUTUREDUET: DECOUPLING OBSERVATION ACCESS FROM FUTURE SUPERVISION IN WORLD ACTION MODELS

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ABSTRACT

World Action Models (WAMs) augment robot action generation with future visual supervision. Existing WAMs commonly fuse main and wrist observations into one visual stream and train both with the same future-video objective, despite their different visual dynamics. A stable main camera reveals scene-level task evolution, whereas wrist cameras move with the end effector, mixing local interaction changes with viewpoint shifts and self-occlusion. These contrasting predictive demands suggest that the two views may benefit from different future objectives. We introduce FutureDuet, which retains both views for control, while allowing each visual stream to receive a different future objective. For the main view, future RGB models task evolution, while interaction masks and robot skeletons focus supervision on task objects and robot motion. For the wrist stream, future latent prediction models short-horizon interaction changes without requiring pixel-level reconstruction. ActionDiT jointly reads the resulting Task State and Interaction State, combining scene-level progress with close-range interaction evidence. All auxiliary prediction modules are training-only, adding no inference overhead. FutureDuet achieves 94.2% clean and 94.1% randomized success on RoboTwin50 and 99.2% average success on LIBERO. The improvements are most pronounced on six RoboTwin50 tasks that require precise interaction, averaging gains of 9.2% and 12.8% over Fast-WAM in clean and randomized settings. Controlled studies further show complementary gains from separating the wrist pathway and designing future supervision separately for the two views.

1 INTRODUCTION

World Action Models (WAMs) augment robot action generation with future visual supervision, leveraging predicted scene evolution to improve policy learning (Bi et al., 2025; Yuan et al., 2026b; Cen et al., 2025b; Kim et al., 2026; Ma et al., 2026; Cheang et al., 2024; Cen et al., 2025a; Zhang et al., 2025; Pai et al., 2025). Fast-WAM further shows that this benefit remains largely when future observations are not generated at deployment (Yuan et al., 2026b). This finding shifts attention from the generation of futures at test time to how future supervision should be designed during training.¹

For WAMs that use both main and wrist cameras, a common design is to fuse their observations into a shared visual stream and train both views under a uniform future-video objective, as illustrated in Fig. 1(a) (Bi et al., 2025; Yuan et al., 2026b; Cen et al., 2025b). This makes incorporating both views straightforward, but entangles observation access with future supervision: every camera stream available to the policy is forced to predict the same form of future target.

However, the camera’s frame of reference changes the prediction problem itself. A main camera provides a relatively stable scene frame in which object relations, robot motion, and task progress

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¹For broader system-level perspectives on integrating AI with sensing, communication, and control, see the AI Flow framework (An et al., 2026; Shao & Li, 2026).

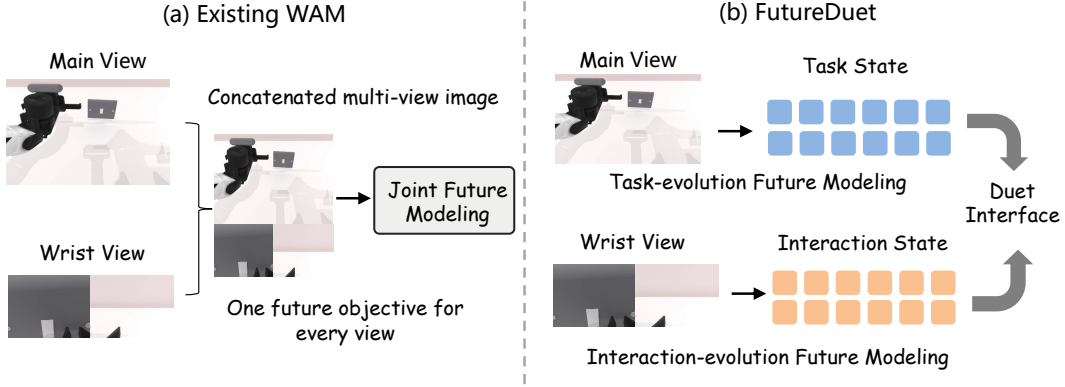


Figure 1: **Decoupling future supervision across views.** (a) A common multi-view WAM design concatenates observations and applies one future objective across views. (b) FutureDuet separately models task evolution from the main view and interaction evolution from wrist views; the resulting states jointly support action generation.

can be tracked over time. Wrist cameras instead move with the end effector, bringing relative pose, contact, and object response into close view while also introducing rapid viewpoint shifts and self-occlusion. From the main view, future changes reveal how the task unfolds across the scene; in wrist views, they reflect both the local interaction and the motion of the camera itself. The two views therefore support complementary state representations for control: one tracks scene-level task progress, while the other captures the ongoing close-range interaction. A shared future video objective nevertheless treats these different visual changes as the same form of prediction target, raising a central question: should each view model the same kind of future?

We therefore introduce FutureDuet (Fig. 1(b)), a WAM that keeps both main and wrist observations available for action generation while allowing their future supervision to be designed separately. For the main view, future RGB prediction captures how the task evolves across the scene, while interaction masks and robot skeletons focus the representation on task objects and robot motion. For the wrist views, future latent prediction models short-horizon interaction changes without requiring pixel-level reconstruction of appearance changes caused by camera motion and self-occlusion. The two streams consequently provide complementary control information: the *Task State* tracks scene-level task progress, while the *Interaction State* captures close-range interaction changes. ActionDiT reads both states jointly to generate actions. All auxiliary prediction modules operate exclusively during training, adding no inference overhead.

FutureDuet achieves 94.2% clean and 94.1% randomized success on RoboTwin50 and 99.2% average success on LIBERO, with its largest RoboTwin50 gains on tasks requiring precise interaction completion. Controlled ablations show gains from wrist-pathway separation; with the deployed architecture fixed, future supervision provides further gains and latent targets perform best. Structured main-view supervision provides complementary gains, while readout diagnostics connect masks and skeletons to object- and robot-related evidence.

Our contributions are:

- We propose FutureDuet, a WAM that decouples observation access from future supervision. It retains main and wrist observations for action generation while allowing their future objectives to be designed separately. The auxiliary prediction modules operate only during training, adding no inference overhead.
- We design two complementary streams for joint control. The main stream learns a *Task State* through future RGB prediction, with masks and skeletons emphasizing task objects and robot motion. The wrist stream learns an *Interaction State* through future latent prediction to capture close-range interaction changes.
- We validate FutureDuet on RoboTwin50 and LIBERO, with the largest RoboTwin50 gains on tasks requiring precise interaction completion. Our controlled wrist-pathway study

shows a 13.9-percentage-point improvement over unified RGB modeling, highlighting the benefits of separate wrist modeling and future supervision.

2 RELATED WORK

Vision-Language-Action Policies and World Action Models. Vision-Language-Action policies map visual observations and language instructions to robot actions, drawing on pretrained vision-language representations and diverse robot demonstrations (Brohan et al., 2023; Octo Model Team et al., 2024; Kim et al., 2025b; Black et al., 2025; NVIDIA et al., 2025; Qu et al., 2025; Liu et al., 2025; Song et al., 2025; Wu et al., 2026). World Action Models augment this mapping with visual future modeling, allowing future observations or representations to shape action learning (Du et al., 2023; Zhou et al., 2024; Bharadhwaj et al., 2025; Feng et al., 2025; Jang et al., 2025). Existing methods use this predictive component in different ways. Some generate future observations as intermediate plans for action prediction (Du et al., 2023; Zhou et al., 2024; Bharadhwaj et al., 2025; Liang et al., 2025; Wu et al., 2023), while others jointly model video and action trajectories (Bi et al., 2025; Cen et al., 2025b; Zhu et al., 2025; Won et al., 2025). Fast-WAM further shows that future prediction can serve as an effective training signal without generating future observations during deployment (Yuan et al., 2026b). Future modeling in WAMs therefore serves not only as an inference mechanism, but also as a source of supervision for learning robot policies.

Predictive Targets for Robot Control. The choice of future target determines which aspects of change a policy is trained to model. RGB targets describe future scene appearance and motion (Hu et al., 2025; Yuan et al., 2026b), whereas latent and joint-embedding objectives model future visual representations without reconstructing every pixel (Sun et al., 2026; Zhang et al., 2026). Recent WAMs use compact latent futures or pretrained joint-embedding spaces to capture predictive dynamics (Chen et al., 2026a; Huang et al., 2026; Lin et al., 2026). Concurrent work W2-VLA predicts task-conditioned future wrist latents and exposes the resulting future-aware context to action generation (Pan et al., 2026). Other approaches emphasize action-relevant structure through object masks, point trajectories, or complementary semantic and geometric targets (Yu et al., 2026; Bharadhwaj et al., 2024; Halder & Pinto, 2025; Yuan et al., 2026a). FutureDuet uses wrist-latent prediction only as training supervision and examines a broader design question: how different future objectives can be assigned across main and wrist streams that are jointly used for action generation.

3 METHOD

3.1 OVERVIEW

At control step t , FutureDuet accepts a language instruction c , proprioception p_t , main-view history $o_{\leq t}^m$, and wrist histories $o_{\leq t}^w$ to generate an action chunk $a = a_{t:t+H-1}$. Both views support control, but their future objectives are designed separately: the main stream models scene-level task evolution, while the wrist stream models short-horizon interaction changes.

As shown in Fig. 2, VideoDiT extracts main-view features $H_t^m = F_m(o_{\leq t}^m, c, p_t)$, and a lightweight online encoder extracts wrist features $H_t^w = E_w(o_{\leq t}^w)$. Learned-query adapters summarize these features into a *Task State* and an *Interaction State*:

$$\begin{aligned} S_t^{\text{task}} &= A_{\text{task}}(Q^{\text{task}}, H_t^m) \in \mathbb{R}^{N_{\text{task}} \times d}, \\ S_t^{\text{int}} &= A_{\text{int}}(Q^{\text{int}}, H_t^w) \in \mathbb{R}^{N_{\text{int}} \times d}. \end{aligned} \quad (1)$$

Each adapter uses learned queries Q to attend to its stream and produce N_{task} or N_{int} tokens of width d . The Task State carries scene context from the instruction and proprioception, while the Interaction State summarizes close-range wrist observations.

The Duet Interface projects and fuses the two states for ActionDiT, which also reads current main-view key/value features K_t^m, V_t^m (Yuan et al., 2026b; Wan et al., 2025; Liang et al., 2024):

$$\begin{aligned} C_t^{\text{duet}} &= A_{\text{duet}}([S_t^{\text{task}}, S_t^{\text{int}}]), \\ \hat{v}_\tau^a &= \Pi_\eta(a^\tau, \tau; C_t^{\text{duet}}, K_t^m, V_t^m, c, p_t). \end{aligned} \quad (2)$$

Rather than fusing main and wrist observations at early visual stages, FutureDuet defers their combination until each stream has been specialized and summarized into its dedicated state. The Duet

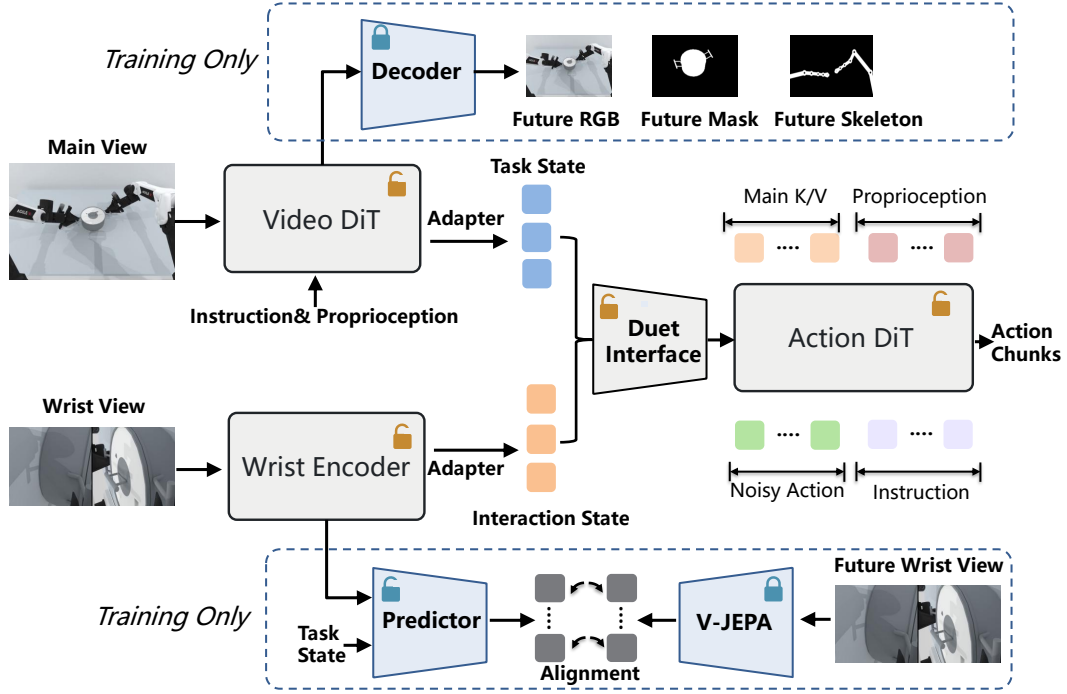


Figure 2: **Overview of FutureDuet.** Different training-only future objectives shape the Task State and Interaction State. ActionDiT reads both through the Duet Interface; auxiliary predictors are removed at deployment.

Interface projects the Task State and Interaction State into a common conditioning space and supplies the fused tokens to ActionDiT. This interface brings the separately supervised representations together for joint action generation. Here $[\cdot; \cdot]$ denotes token-wise concatenation, and a^τ is a noisy action chunk at flow time τ . ActionDiT generates actions by integrating its predicted velocity from noise. This joint readout combines scene-level task context with local interaction evidence.

3.2 TASK STATE AND INTERACTION STATE

Task State from the main view. Future RGB prediction models scene evolution, while interaction masks and robot skeletons focus supervision on task objects and articulated robot motion. The main stream retains the underlying WAM objective: future frames are encoded into VAE latents z^m , and VideoDiT learns to predict their flow velocity under Gaussian noise.

A shared DPT-style decoder reads noise-conditioned intermediate VideoDiT features $\tilde{H}_t^{m,\tau}$. Separate output heads predict interaction masks and skeleton maps at sampled future offsets δ_k . Through the shared decoder, both auxiliary losses supervise these features, emphasizing object extent and robot configuration alongside the appearance information captured by future RGB prediction. Offline SAM 3 (Carion et al., 2025) masks mark the manipulated object and, when required, its target region; skeleton maps project recorded robot configurations into the calibrated main view. All targets are cached offline. The decoder and output heads are trainable and used only during training; App. A.1 provides details on interaction-mask generation and robot-skeleton projection.

Interaction State from the wrist views. Wrist motion mixes local interaction changes with camera-induced appearance variation. We use future latent prediction to supervise temporal structure without requiring pixel reconstruction. A frozen V-JEPA 2.1 (Mur-Labadia et al., 2026) encoder processes future wrist clips offline, producing cached targets $Z_{t,\text{fut}}^w$.

Wrist histories provide local evidence but may not reveal the task context governing the next interaction. The Task State supplies this context to a shared training-only predictor:

$$\hat{Z}_{t,\text{fut}}^w = P_\psi(H_t^w, \text{sg}[S_t^{\text{task}}]). \quad (3)$$

The predictor jointly predicts left- and right-wrist latents, using view embeddings to distinguish the cameras. Future actions are outputs of the policy, not inputs to this predictor. The stop-gradient operator sg prevents the wrist objective from updating the main stream through the conditioning connection, while allowing it to train the online wrist encoder through P_ψ . We minimize the mean absolute error $\mathcal{L}_{\text{wrist}} = |\Omega|^{-1} \sum_{i \in \Omega} |\hat{Z}_{t,\text{fut},i}^w - Z_{t,\text{fut},i}^w|$, where Ω indexes valid latent elements across the batch and both views, excluding padded future target positions.

3.3 JOINT ACTION READOUT AND TRAINING

Main view RGB prediction and action generation use flow matching. Their noisy inputs are $z^{m,\tau} = (1 - \tau)z^m + \tau\epsilon^m$ and $a^\tau = (1 - \tau)a + \tau\epsilon^a$, with independent standard Gaussian noise. The corresponding objectives are

$$\begin{aligned} \mathcal{L}_{\text{rgb}}^m &= \mathbb{E}[w(\tau) \|\hat{v}_\tau^m - (\epsilon^m - z^m)\|_2^2], \\ \mathcal{L}_{\text{act}} &= \mathbb{E}[w(\tau) \|\hat{v}_\tau^a - (\epsilon^a - a)\|_2^2]. \end{aligned} \quad (4)$$

Here $\hat{v}_\tau^m$ and $\hat{v}_\tau^a$ are the VideoDiT and ActionDiT velocity predictions. The two objectives use their respective noise-level samples and scheduler weights, denoted by $\tau \in [0, 1]$ and $w(\tau)$ for brevity. Mask prediction uses focal-Tversky losses, and skeleton prediction uses focal-Dice losses. We jointly optimize

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{act}} + \lambda_{\text{rgb}} \mathcal{L}_{\text{rgb}}^m + \lambda_{\text{mask}} \mathcal{L}_{\text{mask}} \\ &\quad + \lambda_{\text{ske}} \mathcal{L}_{\text{ske}} + \lambda_{\text{wrist}} \mathcal{L}_{\text{wrist}}. \end{aligned} \quad (5)$$

Main-view future losses train VideoDiT, the wrist objective trains the online wrist encoder, and action loss updates both streams through the Duet Interface. At inference, VideoDiT and the wrist encoder form the two states from observed histories, and ActionDiT reads them jointly as in Eq. 2. The VAE decoder, DPT decoder with prediction heads, V-JEPA teacher, and wrist predictor are pruned during evaluation, ensuring that auxiliary future prediction adds no inference overhead. App. A.1 provides detailed training settings.

4 EXPERIMENTS

Our evaluation first measures the complete model across broad bimanual and single-arm benchmarks, then examines three design questions: **Q1**: Why use a separate wrist pathway? **Q2**: Why design structured prediction targets for the main view? **Q3**: And how does this supervision shape the information read by ActionDiT? We answer these questions with benchmark results, controlled component studies, and frozen-checkpoint diagnostics of attention allocation and feature influence.

4.1 EXPERIMENTAL SETUP

Benchmarks. RoboTwin 2.0 (Chen et al., 2026b) contains 50 bimanual tasks in clean and randomized settings and serves as our primary benchmark. We jointly train one model on 50 clean and 500 randomized demonstrations per task, totaling 27,500 demonstrations. This setup matches Fast-WAM in data, backbone, preprocessing, training budget, checkpoint selection, and evaluation. Published baseline values come from the cited benchmark reports; specifically, the UWM result follows the RoboTwin50 evaluation in X-WAM (Guo et al., 2026). LIBERO (Liu et al., 2023) comprises four ten-task, language-conditioned single-arm suites. We use 50 demonstrations per task to evaluate the same design beyond bimanual manipulation.

Controlled studies. The main-view factorial study uses six tasks with 500 randomized demonstrations per task. A separate three-task wrist-pathway study uses the same number of demonstrations and reports three-seed means over 100 randomized evaluation episodes per task and seed. The wrist study fixes main-view supervision to RGB to distinguish the effects of pathway separation and future

Table 1: LIBERO benchmark comparison across four single-arm suites. Success rates (%); higher is better.

Method	Spatial $\uparrow$	Object $\uparrow$	Goal $\uparrow$	Long $\uparrow$	Avg. $\uparrow$
π_0 (Black et al., 2025)	96.8	98.8	95.8	85.2	94.1
GR00T-N1 (NVIDIA et al., 2025)	94.4	97.6	93.0	90.6	93.9
$\pi_{0.5}$ (Physical Intelligence et al., 2025)	98.6	98.2	98.0	92.4	96.8
OpenVLA-OFT (Kim et al., 2025a)	97.6	98.4	97.9	94.5	97.1
Motus (Bi et al., 2025)	96.8	99.8	96.6	97.6	97.7
Fast-WAM (Yuan et al., 2026b)	98.2	100.0	97.0	95.2	97.6
MaskWAM (Yu et al., 2026)	98.8	100.0	98.2	96.4	98.4
LingBot-VA (Li et al., 2026)	98.5	99.6	97.2	98.5	98.5
FutureDuet	99.2	100.0	99.0	98.4	99.2

Table 2: RoboTwin50 benchmark comparison in clean and randomized settings.

Method	Clean $\uparrow$	Rand. $\uparrow$
π_0 (Black et al., 2025)	65.9	58.4
X-VLA (Zheng et al., 2025)	72.9	72.8
$\pi_{0.5}$ (Physical Intelligence et al., 2025)	82.7	76.8
UWM (Zhu et al., 2025)	81.7	78.6
Motus (Bi et al., 2025)	88.7	87.0
X-WAM (Guo et al., 2026)	89.8	90.7
Fast-WAM (Yuan et al., 2026b)	91.9	91.8
LingBot-VA (Li et al., 2026)	92.9	91.5
FutureDuet	94.2	94.1

prediction. The factorial study then tests whether structured main-view targets provide complementary gains under both unified and separate wrist pathways. The action-readout analysis uses another six-task set. App. A.1.2 provides the task lists, protocols, and training settings. Unless stated otherwise, entries are success rates (SR, %); ranked tables mark the best and second-best distinct values with green and yellow backgrounds.

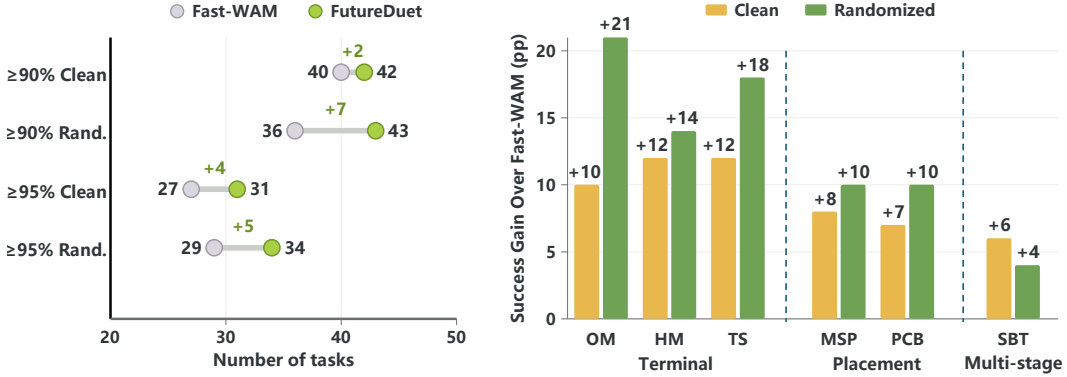


Figure 3: **RoboTwin50 performance varies substantially across tasks.** (a) Number of tasks reaching at least 90% and 95% success under clean and randomized evaluation. (b) Gains over Fast-WAM on the six tasks that remain below 90% for FutureDuet in both settings. Gains are measured in percentage points (pp). OM, HM, TS, MSP, PCB, and SBT denote Open Microwave, Hanging Mug, Turn Switch, Move Stapler Pad, Place Can Basket, and Stack Bowls Three, respectively.

4.2 OVERALL PERFORMANCE

LIBERO. Tab. 1 compares FutureDuet with source-reported results across four single-arm suites. FutureDuet achieves 99.2% average success, with the best reported results on Spatial, Object, and Goal, demonstrating its effectiveness beyond the bimanual RoboTwin setting.

RoboTwin50. FutureDuet achieves 94.2% success in the clean setting and 94.1% under randomization, surpassing the reported Fast-WAM results by 2.3 percentage points in both settings (Tab. 2).

4.3 TASK-LEVEL ANALYSIS ON ROBOTWIN50

Performance across tasks. The high benchmark average coexists with a small set of challenging tasks (Fig. 3). FutureDuet reaches at least 90% success on 41 of 50 tasks in both settings, while six remain below this threshold in both. These tasks require precise terminal interaction, constrained placement, or precision across multiple stages. On this subset, FutureDuet improves over Fast-WAM by 9.2 percentage points in clean evaluation and 12.8 percentage points under randomization, compared with 2.3 points benchmark-wide. Complete results and task groupings appear in App. A.3.

Gains on challenging interactions. The six remaining challenging tasks account for 48% of the total net success gain over Fast-WAM in clean evaluation and 66% under randomization. Open Microwave, Hanging Mug, and Turn Switch show the largest gains within this group in both settings. Their illustrated failures share a common structure: the policy reaches the relevant object but does not complete the final local transition—the door must open far enough, the mug handle must engage the hook, and the end effector must align with and actuate the switch (Fig. 4). These examples suggest that success depends not only on reaching the interaction region, but also on re-solving and executing the contact geometry required for completion. Because wrist views provide close-range evidence of these terminal transitions, these cases motivate the controlled wrist-pathway study in Sec. 4.4, where we isolate the contributions of wrist access, pathway separation, and future supervision on the same three tasks.



Figure 4: **Failure cases on three challenging interaction tasks.** The policy reaches the relevant interaction region but fails to complete the required opening, engagement, or contact.

4.4 WHY USE A SEPARATE WRIST PATHWAY?

We isolate three factors on the wrist side: observation access, pathway separation, and future supervision (Fig. 5). Main-view supervision remains RGB-only throughout this study. We first compare unified and separate wrist pathways, then hold the lightweight pathway fixed to evaluate different prediction targets.

Wrist access and pathway separation. Incorporating wrist observations into the unified pathway boosts mean success from 34.3% to 55.8%, validating the value of close-range wrist cues. Remarkably, a lightweight separate pathway reaches 60.9% even without auxiliary wrist future prediction, demonstrating the intrinsic benefit of a decoupled stream.

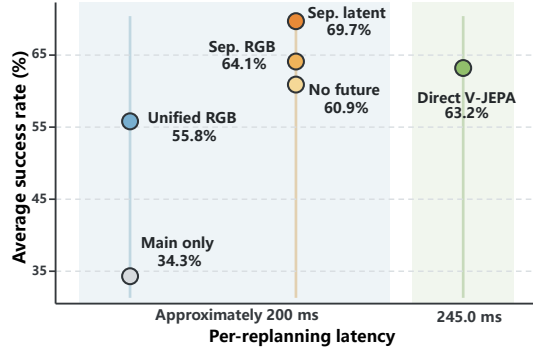


Figure 5: Three-seed wrist-pathway study with RGB-only main-view supervision.

Future supervision under a fixed pathway. Applying future RGB supervision boosts success from 60.9% to 64.1%, whereas future-latent supervision further elevates performance to 69.7%. The RGB result confirms that wrist future prediction is also beneficial, rather than being restricted to latent targets. Nevertheless, the additional 5.6-point gain from latent supervision demonstrates that the choice of target representation is critical within the same pathway. While both objectives provide auxiliary temporal signals beyond action loss, the latent target effectively leverages rich pretrained video features without demanding pixel-level reconstruction. Overall, these findings validate pairing a decoupled wrist pathway with future-latent prediction, achieving optimal performance while preserving deployment efficiency.

Deployment efficiency. Deploying a heavy encoder like V-JEPA highlights the trade-off between model capacity and efficiency, increasing latency from 196.86 ms to 244.99 ms for a 63.2% success rate. We bypass this limitation through training-only auxiliary supervision: confining future-latent prediction strictly to training enables our lightweight pathway to achieve 69.7% success at near-baseline latency (~ 200 ms). Since all auxiliary targets share the same deployed execution graph, upgrading the training target enriches feature representations with zero inference overhead.

Table 3: **Factorial study of main-view supervision and wrist pathways.** The study crosses four main-view target configurations with unified future-RGB and separate future-latent wrist pathways. Fine-grained execution comprises Open Microwave (OM), Hanging Mug (HM), and Turn Switch (TS); additional tasks comprise Adjust Bottle (AB), Blocks Ranking RGB (BR), and Grab Roller (GR). Green and yellow shading indicate the best and second-best result in each column.

Predictive configuration		Fine-grained execution			Additional tasks			Overall
Main-view targets	Wrist pathway	OM	HM	TS	AB	BR	GR	Avg.
RGB	Unified (future RGB)	43	60	57	88	90	88	71.0
RGB+Mask	Unified (future RGB)	43	60	58	92	93	89	72.5
RGB+Skel.	Unified (future RGB)	44	61	59	89	91	92	72.7
RGB+Mask+Skel.	Unified (future RGB)	45	62	59	91	94	90	73.5
RGB	Separate (future latent)	59	66	68	88	88	87	76.0
RGB+Mask	Separate (future latent)	60	66	69	92	93	87	77.8
RGB+Skel.	Separate (future latent)	60	68	70	88	89	90	77.5
RGB+Mask+Skel.	Separate (future latent)	65	70	70	91	94	92	80.3

4.5 DESIGNING STRUCTURED SUPERVISION FOR THE MAIN VIEW

Decoupling future supervision allows us to design the main and wrist pathways around the information they provide for control. With the wrist pathway established, we now investigate supervision strategies for the main view. Future RGB prediction captures scene evolution but does not explicitly prioritize the task objects and robot motion that guide manipulation. We therefore study whether masks and skeletons strengthen this supervision and complement the separate wrist pathway (Tab. 3).

Complementarity with wrist modeling. Adding masks and skeletons raises average success from 71.0% to 73.5% under the unified pathway and from 76.0% to 80.3% under the separate future-latent pathway, with gains spanning both fine-grained execution and the additional tasks. With RGB-only main-view supervision, the gains from the separate future-latent pathway are concentrated on OM, HM, and TS. Adding structured main-view targets also improves AB, BR, and GR under both pathways, broadening the benefits beyond the fine-grained tasks most responsive to wrist modeling. These improvements show that the value of separately designing future supervision extends beyond the wrist pathway. Alongside wrist-side prediction, explicitly emphasizing task objects and robot motion in the main-view objective provides additional gains, with the combined design achieving the highest average success under both wrist pathways.

Distinct contributions of masks and skeletons. Under both wrist pathways, masks provide larger gains than skeletons on Adjust Bottle and Blocks Ranking RGB, whereas skeletons provide larger gains on Grab Roller. Neither target is consistently better across these tasks, and combining them achieves the highest average success under either pathway. This pattern supports supervising both task objects and robot motion rather than relying on either target alone.

Together, these results show that structured main view supervision complements wrist modeling, with masks and skeletons contributing differently across tasks. We next examine whether this functional division is mirrored in how ActionDiT allocates attention to task objects and robot features.

4.6 UNDERSTANDING THE EFFECTS OF STRUCTURED PREDICTION TARGETS

The factorial study shows that masks and skeletons produce different task-level gains. We therefore examine whether they change the main-view evidence used by ActionDiT. Across 48 matched queries from separately trained six-task checkpoints, we first use action-to-visual attention to measure where action queries retrieve information, and then apply matched K/V interventions to test whether features from those regions affect the action output.

Where does ActionDiT read? We extract action-query-to-main-view attention from the final two joint VideoDiT–ActionDiT attention blocks. RR(O) and RR(G) quantify action attention directed

to task-object and executing-robot regions relative to a size-matched background, whereas OCE measures object-center alignment, with lower values indicating better alignment.

The effects are target-specific. Under the separate wrist pathway, mask supervision raises object readout $RR(O)$ from 1.10 to 1.46 and improves object-center alignment, whereas skeleton supervision produces the strongest robot-side change, raising $RR(G)$ from 1.20 to 2.00. The same division appears under unified wrist modeling, indicating that it is induced by the main-view targets rather than the wrist pathway. Fig. 6 qualitatively illustrates this effect, showing how joint supervision concentrates action readout on the task-relevant interaction.

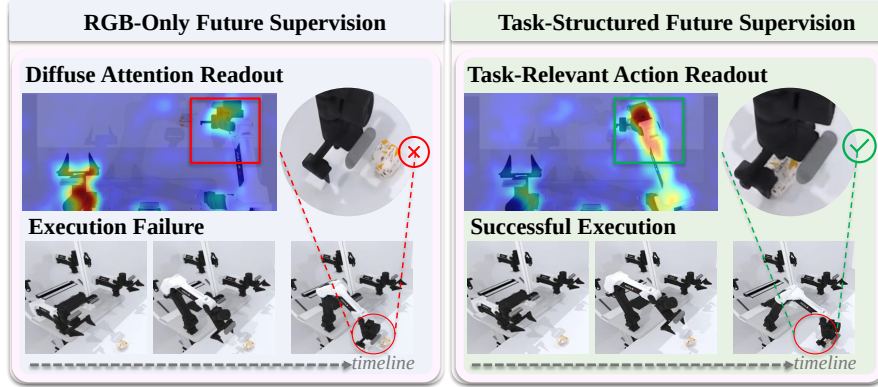


Figure 6: **Structured targets focus action readout.** Compared with RGB-only future supervision, joint mask-and-skeleton supervision focuses readout on the task-relevant interaction.

Which regions influence action generation? Attention indicates where action queries retrieve information, but not whether the retrieved features affect the output. We therefore replace object- or robot-region K/V features with the mean main-view K/V and measure the resulting action change using $R(O)$ and $R(G)$.

The intervention results mirror the attention patterns. Mask supervision primarily increases sensitivity to object features, whereas skeleton supervision primarily increases sensitivity to robot features; joint supervision preserves both effects under either wrist pathway. Thus, the target-specific readout is reflected not only in attention patterns but also in the action model’s response to the corresponding visual features.

	Task-Object Readout			Robot-State Readout	
	$RR(O)$	$R(O)$	$OCE\downarrow$	$RR(G)$	$R(G)$
Mask	+0.36	+0.63	+0.68	+0.04	+0.06
Skel.	+0.03	+0.10	+0.13	+0.80	+0.89
Both	+0.47	+1.00	+0.92	+0.96	+1.17

Figure 7: **Target-specific changes under separate wrist modeling.** Positive values indicate increases in attention readout and feature sensitivity, as well as reductions in OCE.

Together, the two diagnostics provide complementary evidence: attention identifies where Action-DiT reads, while K/V intervention shows that the corresponding features affect its output. Their agreement suggests that mask and skeleton supervision shape complementary sides of the Task State—the task object and the robot executing the motion—helping explain why their combination performs best in the factorial study. Full results appear in Appendix Tab. A10.

5 CONCLUSION

FutureDuet decouples two choices that existing WAMs often entangle together: which observations support control and what future each visual stream is trained to predict. By using main and wrist observations for action generation while predicting structured main-view targets and future wrist latents, our framework learns complementary representations of overall task progress and local interaction. This decoupled design improves performance across RoboTwin50 and LIBERO, with

the largest RoboTwin50 gains appearing on tasks requiring precise interaction completion. Controlled studies isolate two key sources of wrist-side improvement: pathway separation provides a fundamental architectural benefit, while supervision target selection favors future-latent prediction over pixel-level RGB reconstruction under the same deployed architecture. Meanwhile, structured main-view supervision offers complementary gains. While our evaluation is currently limited to simulation, future work will test FutureDuet on real robots and explore broader view-target configurations. Overall, our results demonstrate that distinct camera streams can jointly guide action control without forcing every view to predict a uniform future, highlighting the value of decoupling observation access from training-time auxiliary future supervision in world action models.

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